Compulsory Assignment 2: Convolutional neural networks

Please fill out the group name, number, members and optionally the name below.

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Group name (optional):

Assignment Submission

To complete this assignment answer the relevant questions in this notebook and write the code required to implement the relevant models. The assignment is submitted by handing in this notebook as an .ipynb file and as a .pdf file.

Introduction

In this assignment, you will build a Convolutional Neural Network (CNN) to classify images of natural scenes from around the world. Dataset: Intel-image-classification https://www.kaggle.com/datasets/puneet6060/intel-image-classification

This Data contains around 25k images of size 150x150 distributed under 6 categories. {'buildings' -> 0, 'forest' -> 1, 'glacier' -> 2, 'mountain' -> 3, 'sea' -> 4, 'street' -> 5 }

In CA2, we use only 3 classes "buildings", "forest", "sea".

This data was initially published on https://datahack.analyticsvidhya.com by Intel to host a Image classification Challenge.

Assignment structure

1. Part 0: Setup & Data
2. Part 1: Baseline CNN (Clean Data)
3. Part 2: Choose Your Robustness Challenges
4. Part 3: Results & Comparison

```
## Library imports
```

```
# Feel free to add or remove libraries as you want
import time
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
import tensorflow.keras as ks
import sklearn
from sklearn.metrics import accuracy_score
```

Part 0: Setup & Data, CNN for Landscape Picture dataset

NOTE FOR STUDENTS:

Do not forget to change the path of root_dir below so it points to the Intel folder which contains the dataset inside your CA2 directory.

Example:

Windows: r"C:\Users\yourname...\CA2\Intel"

Mac/Linux: "/home/yourname/CA2/Intel"

Loading DATASET

```
import importlib
import utilities

# Force reload the updated utilities.py
importlib.reload(utilities)

from utilities import load_intel_dataset


# Load dataset
data = load_intel_dataset(
    root_dir="C:\\Users\\fredr\\OneDrive - Norwegian University of Life Sciences\\DAT300\\CA\\CA_2\\Intel",
    img_size=(224, 224),
    selected_classes=["buildings", "forest", "sea"],  #  only 3 classes
    verbose=1

)

X_train, y_train = data["X_train"], data["y_train"]
X_val,   y_val   = data["X_val"],   data["y_val"]
X_test,  y_test  = data["X_test"],  data["y_test"]
class_names      = data["class_names"]

print("Train:", X_train.shape, y_train.shape)
print("Val:",   X_val.shape,   y_val.shape)
print("Test:",  X_test.shape,  y_test.shape)

# Show number of images per class
def count_per_class(y, split_name):
    unique, counts = np.unique(y, return_counts=True)
    print(f"\n{split_name} set class distribution:")
    for cls, cnt in zip(unique, counts):
        print(f"  {class_names[cls]}: {cnt}")

count_per_class(y_train, "Train")
count_per_class(y_val,   "Val")
count_per_class(y_test,  "Test")
```

```
[train] Loaded 500/6736 images
[train] Loaded 1000/6736 images
[train] Loaded 1500/6736 images
[train] Loaded 2000/6736 images
[train] Loaded 2500/6736 images
[train] Loaded 3000/6736 images
[train] Loaded 3500/6736 images
[train] Loaded 4000/6736 images
[train] Loaded 4500/6736 images
[train] Loaded 5000/6736 images
[train] Loaded 5500/6736 images
[train] Loaded 6000/6736 images
[train] Loaded 6500/6736 images
[train] Loaded 6736/6736 images
[test] Loaded 500/1156 images
[test] Loaded 1000/1156 images
[test] Loaded 1156/1156 images
Final shapes → Train: (6062, 224, 224, 3), Val: (674, 224, 224, 3), Test: (1156, 224, 224, 3)
Classes: ['buildings', 'forest', 'sea']
Train: (6062, 224, 224, 3) (6062,)
Val: (674, 224, 224, 3) (674,)
Test: (1156, 224, 224, 3) (1156,)

Train set class distribution:
  buildings: 1972
  forest: 2044
  sea: 2046

Val set class distribution:
  buildings: 219
  forest: 227
```

```
  sea: 228

Test set class distribution:
  buildings: 437
  forest: 474
  sea: 245
```

Preprocessing

```
# Normalizing input between [0,1]
X_train = X_train.astype("float32") / 255.0 if X_train.size else X_train
X_val   = X_val.astype("float32")   / 255.0 if X_val.size else X_val
X_test  = X_test.astype("float32")  / 255.0 if X_test.size else X_test

# Converting targets from numbers to categorical format
if y_train is not None and y_train.size:
    num_classes = len(np.unique(y_train))
    y_train = ks.utils.to_categorical(y_train, num_classes)

if y_val is not None and y_val.size:
    y_val = ks.utils.to_categorical(y_val, num_classes)

if y_test is not None and y_test.size:
    y_test = ks.utils.to_categorical(y_test, num_classes)

print("X_train shape:", X_train.shape)
print("X_val shape:",   X_val.shape)
print("X_test shape:",  X_test.shape)
```

```
X_train shape: (6062, 224, 224, 3)
X_val shape: (674, 224, 224, 3)
X_test shape: (1156, 224, 224, 3)
```

Part 1: Build a CNN network with the LeNet5 architecture

Implement LeNet5 architecture for Landscape Pictures (RGB):

The LeNet architecture takes a 32×32×C image as input, where C is the number of color channels. You may use resize_dataset() and f1_score() from utilities.py¶

Input & resizing: Resize all images to 32×32 and use C = 3 channels (RGB). If you choose a different input size, update the intermediate shapes accordingly, but keep the LeNet-5 pattern (Conv → Pool → Conv → Pool → FC → FC → FC).

Layer 1 - Convolution (5x5): The output shape should be 28x28x6. **Activation:** ReLU.

MaxPooling: The output shape should be 14x14x6.

Layer 2 - Convolution (5x5): The output shape should be 10x10x16. **Activation:** ReLU.

MaxPooling: The output shape should be 5x5x16.

Flatten: Flatten the output shape of the final pooling layer such that it's 1D instead of 3D. You may need to use tf.reshape.

Layer 3 - Fully Connected: This should have 120 outputs. **Activation:** ReLU.

Layer 4 - Fully Connected: This should have 84 outputs. **Activation:** ReLU.

Layer 5 - Fully Connected (output): `num_classes`. **Activation:** softmax

---

Compile the network with the

- `tf.keras.losses.CategoricalCrossentropy` loss function
- the `adam` optimizer
- with the `accuracy` metric and (your own implementation of the) F1-score metric.

```
X_train = utilities.resize_dataset(X_train)
X_val = utilities.resize_dataset(X_val)
X_test = utilities.resize_dataset(X_test)
print(X_train.shape)
print(X_val.shape)
print(X_test.shape)
```

```
(6062, 32, 32, 3)
(674, 32, 32, 3)
(1156, 32, 32, 3)
```

```
ks.utils.set_random_seed(42)
model = ks.Sequential([
    ks.Input(shape=(32, 32, 3)),
    ks.layers.Conv2D(filters=6, kernel_size=(5), activation="relu"),
    ks.layers.MaxPool2D(pool_size=(2, 2)),
    ks.layers.Conv2D(filters=16, kernel_size=(5), activation="relu"),
    ks.layers.MaxPool2D(pool_size=(2, 2)),
    ks.layers.Flatten(),
    ks.layers.Dense(120, activation="relu"),
    ks.layers.Dense(84, activation="relu"),
    ks.layers.Dense(3, activation="softmax")
    #ks.layers.Dense(128, activation="sigmoid", activity_regularizer=keras.regularizers.L2(1e-5)),
    #ks.layers.Dense(64, activation="sigmoid", activity_regularizer=keras.regularizers.L2(1e-5)),
    #ks.layers.Dense(24, activation="softmax", activity_regularizer=keras.regularizers.L2(1e-5))
])

model.summary()

optimizer = ks.optimizers.Adam()

# Compile model
model.compile(optimizer=optimizer,
              loss='categorical_crossentropy',
              metrics=['accuracy']
              )
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
conv2d_4 (Conv2D)	(None, 28, 28, 6)	456
max_pooling2d_4 (MaxPooling2D)	(None, 14, 14, 6)	0
conv2d_5 (Conv2D)	(None, 10, 10, 16)	2,416
max_pooling2d_5 (MaxPooling2D)	(None, 5, 5, 16)	0
flatten_3 (Flatten)	(None, 400)	0
dense_11 (Dense)	(None, 120)	48,120
dense_12 (Dense)	(None, 84)	10,164
dense_13 (Dense)	(None, 3)	255

Total params: 61,411 (239.89 KB)
Trainable params: 61,411 (239.89 KB)

Task 1.1.2 Train network

Train the network with a

- batch size of 64 samples
- for 20 epochs

```
filepath_model = "model_best.weights.h5"
model_checkpoint_callback = ks.callbacks.ModelCheckpoint(
        filepath=filepath_model,
        save_weights_only=True,
        monitor='val_accuracy',
        mode='max',
```

```
        save_best_only=True
    )

training_history_object = model.fit(X_train,
          y_train,
          validation_data=(X_val, y_val),
          epochs=20,
          batch_size=64,
          callbacks = [model_checkpoint_callback],
          )

model.load_weights(filepath_model)

accuracy = max(model.history.history["val_accuracy"])
f1_score = ks.backend.get_value(utilities.f1_score(y_val, model.predict(X_val)))

print("\nModel:")
print(f"- Validation f1: {f1_score*100:.2f}%")
print(f"- Validation Accuracy: {accuracy*100:.2f}%")
```

```
Epoch 1/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 4s 20ms/step - accuracy: 0.5318 - loss: 0.9776 - val_accuracy: 0.5742 - val_loss: 0.899
Epoch 2/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.6198 - loss: 0.8079 - val_accuracy: 0.6528 - val_loss: 0.734
Epoch 3/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6353 - loss: 0.7823 - val_accuracy: 0.6751 - val_loss: 0.718
Epoch 4/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6420 - loss: 0.7679 - val_accuracy: 0.6780 - val_loss: 0.706
Epoch 5/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6509 - loss: 0.7548 - val_accuracy: 0.6780 - val_loss: 0.702
Epoch 6/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6567 - loss: 0.7417 - val_accuracy: 0.6780 - val_loss: 0.705
Epoch 7/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6679 - loss: 0.7284 - val_accuracy: 0.6751 - val_loss: 0.714
Epoch 8/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6740 - loss: 0.7143 - val_accuracy: 0.6647 - val_loss: 0.722
Epoch 9/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6838 - loss: 0.6972 - val_accuracy: 0.6513 - val_loss: 0.740
Epoch 10/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.6984 - loss: 0.6727 - val_accuracy: 0.6543 - val_loss: 0.741
Epoch 11/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.7273 - loss: 0.6284 - val_accuracy: 0.6766 - val_loss: 0.704
Epoch 12/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.7461 - loss: 0.5951 - val_accuracy: 0.6914 - val_loss: 0.667
Epoch 13/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 1s 15ms/step - accuracy: 0.7615 - loss: 0.5597 - val_accuracy: 0.7404 - val_loss: 0.598
Epoch 14/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 15ms/step - accuracy: 0.7836 - loss: 0.5237 - val_accuracy: 0.7463 - val_loss: 0.586
Epoch 15/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.7923 - loss: 0.5002 - val_accuracy: 0.7700 - val_loss: 0.546
Epoch 16/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.8052 - loss: 0.4841 - val_accuracy: 0.8056 - val_loss: 0.479
Epoch 17/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.8105 - loss: 0.4691 - val_accuracy: 0.8116 - val_loss: 0.463
Epoch 18/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.8159 - loss: 0.4594 - val_accuracy: 0.8131 - val_loss: 0.456
Epoch 19/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.8179 - loss: 0.4533 - val_accuracy: 0.8234 - val_loss: 0.451
Epoch 20/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 16ms/step - accuracy: 0.8192 - loss: 0.4462 - val_accuracy: 0.8249 - val_loss: 0.446
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step

Model:
- Validation f1: 90.38%
- Validation Accuracy: 82.49%
```

Task 1.1.3

Experiment with different architectures of your choice. Vary the number of filters, try different kernel sizes, add more layers, dropout, early stopping, and modify the fully connected layers. Report the best performance you achieve.

```
model2 = ks.Sequential([
        ks.Input(shape=(32, 32, 3)),
        ks.layers.Conv2D(filters=10, kernel_size=(9), activation="relu"),
        ks.layers.MaxPool2D(pool_size=(4, 4)),
        ks.layers.Flatten(),
        ks.layers.Dense(120, activation="relu"),
        ks.layers.Dense(84, activation="sigmoid"),
        ks.layers.Dense(32, activation="relu"),
        ks.layers.Dense(3, activation="softmax")
    ])

model2.summary()

optimizer = ks.optimizers.Adam()

# Compile model
model2.compile(optimizer=optimizer,
              loss='categorical_crossentropy',
              metrics=['accuracy']
              )


filepath_model2 = "model2_best.weights.h5"
model_checkpoint_callback_2 = ks.callbacks.ModelCheckpoint(
        filepath=filepath_model2,
        save_weights_only=True,
        monitor='val_accuracy',
        mode='max',
        save_best_only=True
    )

training_history_object_2 = model2.fit(X_train,
          y_train,
          validation_data=(X_val, y_val),
          epochs=20,
          batch_size=64,
          callbacks = [model_checkpoint_callback_2]
          )

model2.load_weights(filepath_model2)

accuracy_2 = max(model2.history.history["val_accuracy"])
f1_score_2 = ks.backend.get_value(utilities.f1_score(y_val, model2.predict(X_val)))

print("\nModel 2:")
print(f"- Validation f1: {f1_score_2*100:.2f}%")
print(f"- Validation accuracy: {accuracy_2*100:.2f}%")
```

```
Model: "sequential_4"
```

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 24, 24, 10)	2,440
max_pooling2d_6 (MaxPooling2D)	(None, 6, 6, 10)	0
flatten_4 (Flatten)	(None, 360)	0
dense_14 (Dense)	(None, 120)	43,320
dense_15 (Dense)	(None, 84)	10,164
dense_16 (Dense)	(None, 32)	2,720
dense_17 (Dense)	(None, 3)	99

```
Total params: 58,743 (229.46 KB)
Trainable params: 58,743 (229.46 KB)
Non-trainable params: 0 (0.00 B)
Epoch 1/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 4s 21ms/step - accuracy: 0.4296 - loss: 1.0608 - val_accuracy: 0.6231 - val_loss: 0.858
Epoch 2/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.5896 - loss: 0.8159 - val_accuracy: 0.6380 - val_loss: 0.749
Epoch 3/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.6110 - loss: 0.7904 - val_accuracy: 0.6454 - val_loss: 0.735
Epoch 4/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step - accuracy: 0.6211 - loss: 0.7771 - val_accuracy: 0.6513 - val_loss: 0.723
Epoch 5/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.6298 - loss: 0.7643 - val_accuracy: 0.6558 - val_loss: 0.710
Epoch 6/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.6415 - loss: 0.7508 - val_accuracy: 0.6602 - val_loss: 0.697
Epoch 7/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.6521 - loss: 0.7358 - val_accuracy: 0.6721 - val_loss: 0.683
Epoch 8/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step - accuracy: 0.6663 - loss: 0.7175 - val_accuracy: 0.6884 - val_loss: 0.666
Epoch 9/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step - accuracy: 0.6834 - loss: 0.6950 - val_accuracy: 0.7107 - val_loss: 0.645
Epoch 10/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7027 - loss: 0.6684 - val_accuracy: 0.7226 - val_loss: 0.616
Epoch 11/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step - accuracy: 0.7308 - loss: 0.6352 - val_accuracy: 0.7493 - val_loss: 0.578
Epoch 12/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7468 - loss: 0.6020 - val_accuracy: 0.7730 - val_loss: 0.541
Epoch 13/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7646 - loss: 0.5711 - val_accuracy: 0.7864 - val_loss: 0.510
Epoch 14/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7783 - loss: 0.5454 - val_accuracy: 0.8056 - val_loss: 0.488
Epoch 15/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7879 - loss: 0.5244 - val_accuracy: 0.8145 - val_loss: 0.473
Epoch 16/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 19ms/step - accuracy: 0.7953 - loss: 0.5065 - val_accuracy: 0.8190 - val_loss: 0.459
Epoch 17/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 22ms/step - accuracy: 0.8039 - loss: 0.4907 - val_accuracy: 0.8234 - val_loss: 0.443
Epoch 18/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 20ms/step - accuracy: 0.8114 - loss: 0.4758 - val_accuracy: 0.8249 - val_loss: 0.429
Epoch 19/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.8141 - loss: 0.4634 - val_accuracy: 0.8294 - val_loss: 0.417
Epoch 20/20
95/95 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step - accuracy: 0.8213 - loss: 0.4518 - val_accuracy: 0.8383 - val_loss: 0.406
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step

Model 2:
- Validation f1: 92.65%
```

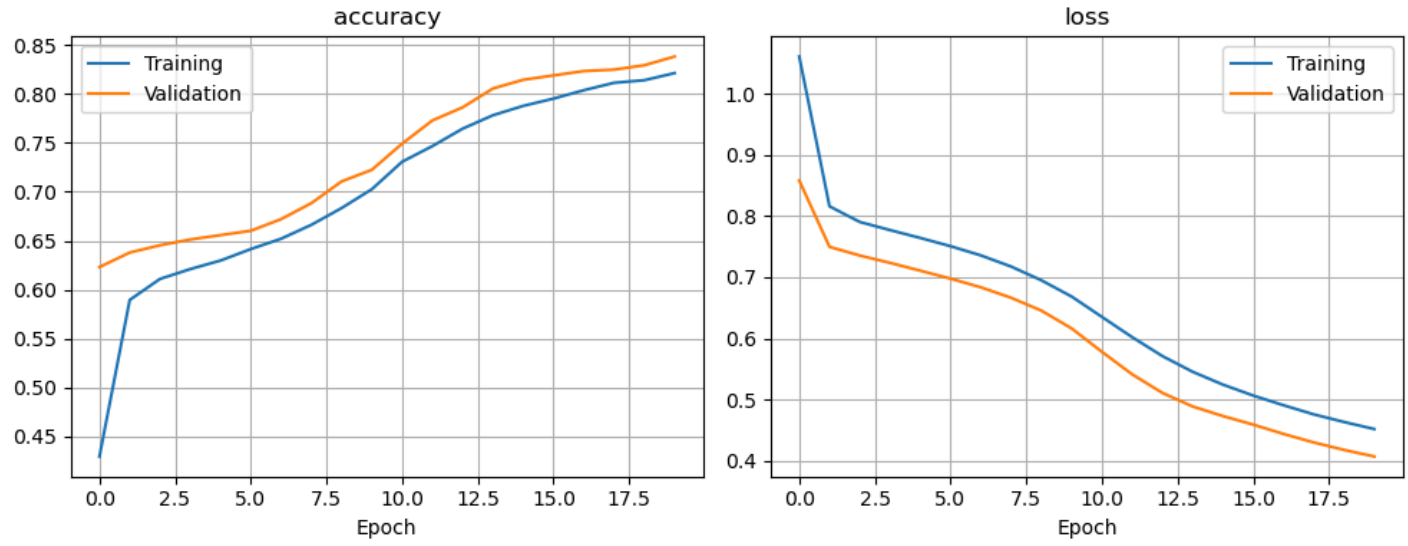
Dobbeltklikk (eller trykk på Enter) for å redigere

Task 1.2 Evaluaiton

Task 1.2.1 Plot training history

- Plot the training/validation accuracy and loss curves (plot_training_history() is in utilities.py).
- Report the final validation accuracy (f1_score() is in utilities.py).

```
utilities.plot_training_history(training_history_object_2, ['accuracy', 'loss'])
```



Task 1.2.2 Evaluate on the test dataset

- Report test accuracy (and F1 score (f1_score() is in utilities.py)).

```
# Make predictions
y_pred_2 = model2.predict(X_test)

print('Artificial neural network:')

# Calculate f1
f1_2 = utilities.f1_score(y_test, y_pred_2)
print(f'- Classification f1: {f1_2 * 100:.2f}%')

# Calculate accuracy
accuracy_2 = accuracy_score(y_test.argmax(axis=1), y_pred_2.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_2 * 100:.2f}%')
```

```
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
Artificial neural network:
- Classification f1: 89.66%
- Classification Accuracy: 82.27%
```

Task 1.2.3 Create a confusion matrix for both training and testing data

- Visualize confusion matrices.
- Do the test data and train data predict the same items wrong?

```
conf_mat_train = sklearn.metrics.confusion_matrix(y_train.argmax(axis=1), model2.predict(X_train).argmax(axis=1))
train_instances = np.sum(conf_mat_train, axis=1)
# print(conf_mat_train)
# print(train_instances)

print("Scaled train confusion matrix")
print(np.transpose(conf_mat_train) / train_instances)



conf_mat_test = sklearn.metrics.confusion_matrix(y_test.argmax(axis=1), y_pred_2.argmax(axis=1))
test_instances = np.sum(conf_mat_test, axis=1)
print("\nScaled test confusion matrix")
print(np.transpose(conf_mat_test) / test_instances)
```

```
190/190 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step
Scaled train confusion matrix
[[0.74391481 0.08610568 0.12072336]
 [0.09837728 0.8909002  0.02101662]
```

```
 [0.15770791 0.02299413 0.85826002]]

Scaled test confusion matrix
[[0.74828375 0.10759494 0.11020408]
 [0.1006865  0.86919831 0.0244898 ]
 [0.15102975 0.02320675 0.86530612]]
```

There are no obvious differences between the confusion matrices. This indicates that the model makes the same kind of prediction errors for both the training and test data.

Part 2: Robustness (choose two or more), You may use add_gaussian_noise() and add_motion_blur() from utilities.py

Task 2.1 Data Augmentation (training-time)

- Implement an augmentation pipeline.
- Train a model with augmentation and compare against the baseline from Part 1.

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
datagen = ImageDataGenerator(width_shift_range=0.2,
                             height_shift_range=0.2,
                             zoom_range=0.2,
                             rotation_range=10,
                             vertical_flip=False,
                             horizontal_flip=False)
datagen.fit(X_train)
train_generator = datagen.flow(X_train, y_train,
                               batch_size=32)
```

```
from tensorflow.keras.models import clone_model
model3 = clone_model(model2)

#model3.fit(train_generator, validation_data=(X_val, y_val), epochs=4, batch_size=32)

model3.summary()

optimizer = ks.optimizers.Adam()

# Compile model
model3.compile(optimizer=optimizer,
               loss='categorical_crossentropy',
               metrics=['accuracy']
               )


filepath_model3 = "model3_best.weights.h5"
model_checkpoint_callback_3 = ks.callbacks.ModelCheckpoint(
        filepath=filepath_model3,
        save_weights_only=True,
        monitor='val_accuracy',
        mode='max',
        save_best_only=True
    )

training_history_object_3 = model3.fit(train_generator,
          validation_data=(X_val, y_val),
          epochs=20,
          batch_size=64,
          callbacks = [model_checkpoint_callback_3]
          )

model3.load_weights(filepath_model3)

accuracy_3 = max(model3.history.history["val_accuracy"])
f1_score_3 = ks.backend.get_value(utilities.f1_score(y_val, model3.predict(X_val)))
```

```
print("\nModel 3:")
print(f"- Validation f1: {f1_score_3*100:.2f}%")
print(f"- Validation Accuracy: {accuracy_3*100:.2f}%")
```

Model: "sequential_4"

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 24, 24, 10)	2,440
max_pooling2d_6 (MaxPooling2D)	(None, 6, 6, 10)	0
flatten_4 (Flatten)	(None, 360)	0
dense_14 (Dense)	(None, 120)	43,320
dense_15 (Dense)	(None, 84)	10,164
dense_16 (Dense)	(None, 32)	2,720
dense_17 (Dense)	(None, 3)	99

Total params: 176,231 (688.41 KB)
Trainable params: 58,743 (229.46 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 117,488 (458.94 KB)

```
Epoch 1/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 6s 20ms/step - accuracy: 0.4805 - loss: 0.9953 - val_accuracy: 0.6395 - val_loss: 0.7
Epoch 2/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 4s 18ms/step - accuracy: 0.5876 - loss: 0.8406 - val_accuracy: 0.6409 - val_loss: 0.7
Epoch 3/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.5868 - loss: 0.8192 - val_accuracy: 0.6469 - val_loss: 0.7
Epoch 4/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - accuracy: 0.6104 - loss: 0.8035 - val_accuracy: 0.6217 - val_loss: 0.7
Epoch 5/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6320 - loss: 0.7911 - val_accuracy: 0.6766 - val_loss: 0.7
Epoch 6/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6478 - loss: 0.7765 - val_accuracy: 0.7033 - val_loss: 0.6
Epoch 7/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6476 - loss: 0.7696 - val_accuracy: 0.7107 - val_loss: 0.6
Epoch 8/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6653 - loss: 0.7500 - val_accuracy: 0.7122 - val_loss: 0.6
Epoch 9/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6854 - loss: 0.7252 - val_accuracy: 0.7285 - val_loss: 0.6
Epoch 10/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 17ms/step - accuracy: 0.6826 - loss: 0.7184 - val_accuracy: 0.7270 - val_loss: 0.6
Epoch 11/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.6952 - loss: 0.7070 - val_accuracy: 0.7463 - val_loss: 0.6
Epoch 12/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.7017 - loss: 0.6864 - val_accuracy: 0.7552 - val_loss: 0.6
Epoch 13/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.7115 - loss: 0.6703 - val_accuracy: 0.7507 - val_loss: 0.5
Epoch 14/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.7060 - loss: 0.6757 - val_accuracy: 0.7507 - val_loss: 0.5
Epoch 15/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.7125 - loss: 0.6570 - val_accuracy: 0.7671 - val_loss: 0.5
Epoch 16/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 4s 20ms/step - accuracy: 0.7288 - loss: 0.6369 - val_accuracy: 0.7789 - val_loss: 0.5
Epoch 17/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 4s 18ms/step - accuracy: 0.7291 - loss: 0.6279 - val_accuracy: 0.7700 - val_loss: 0.5
Epoch 18/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 4s 19ms/step - accuracy: 0.7270 - loss: 0.6174 - val_accuracy: 0.7878 - val_loss: 0.5
Epoch 19/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 4s 18ms/step - accuracy: 0.7448 - loss: 0.5987 - val_accuracy: 0.7878 - val_loss: 0.5
Epoch 20/20
190/190 ━━━━━━━━━━━━━━━━━━━━ 3s 18ms/step - accuracy: 0.7465 - loss: 0.5988 - val_accuracy: 0.7849 - val_loss: 0.5
22/22 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step

Model 3:
- Validation f1: 94.57%
```

```
print('Model 2:')
print(f'- Classification f1: {f1_2 * 100:.2f}%')
print(f'- Classification Accuracy: {accuracy_2 * 100:.2f}%')


print('\nModel 3:')
```

```
# Make predictions
y_pred_3 = model3.predict(X_test)

# Calculate f1
f1_3 = utilities.f1_score(y_test, y_pred_3)
print(f'- Classification f1: {f1_3 * 100:.2f}%')

# Calculate accuracy
accuracy_3 = accuracy_score(y_test.argmax(axis=1), y_pred_3.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_3 * 100:.2f}%')
```

```
Model 2:
- Classification f1: 89.66%
- Classification Accuracy: 82.27%

Model 3:
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
- Classification f1: 90.81%
- Classification Accuracy: 83.82%
```

The augmented model performs minimally better.

Task 2.2 Noise Robustness (test-time corruptions)

- Create corrupted versions of the test images and evaluate the baseline model:
  - Gaussian noise
  - Motion blur
- Report accuracy on the clean test set vs. each corrupted test set.

```
X_test_gauss = utilities.add_gaussian_noise(X_test)
X_test_blur = utilities.add_motion_blur(X_test)


print('Model 2:')
print(f'- Classification f1: {f1_2 * 100:.2f}%')
print(f'- Classification Accuracy: {accuracy_2 * 100:.2f}%')


print('\nModel 2 (Gaussian noise):')
y_pred_2_gauss = model2.predict(X_test_gauss)
f1_2_gauss = utilities.f1_score(y_test, y_pred_2_gauss)
print(f'- Classification f1: {f1_2_gauss * 100:.2f}%')
accuracy_2_gauss = accuracy_score(y_test.argmax(axis=1), y_pred_2_gauss.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_2_gauss * 100:.2f}%')

print('\nModel 2 (Motion blur):')
y_pred_2_blur = model2.predict(X_test_blur)
f1_2_blur = utilities.f1_score(y_test, y_pred_2_blur)
print(f'- Classification f1: {f1_2_blur * 100:.2f}%')
accuracy_2_blur = accuracy_score(y_test.argmax(axis=1), y_pred_2_blur.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_2_blur * 100:.2f}%')


print('\n\nModel 3:')
print(f'- Classification f1: {f1_3 * 100:.2f}%')
print(f'- Classification Accuracy: {accuracy_3 * 100:.2f}%')


print('\nModel 3 (Gaussian noise):')
y_pred_3_gauss = model3.predict(X_test_gauss)
f1_3_gauss = utilities.f1_score(y_test, y_pred_3_gauss)
print(f'- Classification f1: {f1_3_gauss * 100:.2f}%')
accuracy_3_gauss = accuracy_score(y_test.argmax(axis=1), y_pred_3_gauss.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_3_gauss * 100:.2f}%')

print('\nModel 3 (Motion blur):')
y_pred_3_blur = model3.predict(X_test_blur)
```

```
f1_3_blur = utilities.f1_score(y_test, y_pred_3_blur)
print(f'- Classification f1: {f1_3_blur * 100:.2f}%')
accuracy_3_blur = accuracy_score(y_test.argmax(axis=1), y_pred_3_blur.argmax(axis=1))
print(f'- Classification Accuracy: {accuracy_3_blur * 100:.2f}%')
```

```
Model 2:
- Classification f1: 89.66%
- Classification Accuracy: 82.27%

Model 2 (Gaussian noise):
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
- Classification f1: 0.00%
- Classification Accuracy: 37.80%

Model 2 (Motion blur):
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step
- Classification f1: 99.18%
- Classification Accuracy: 36.85%


Model 3:
- Classification f1: 90.81%
- Classification Accuracy: 83.82%

Model 3 (Gaussian noise):
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
- Classification f1: 0.00%
- Classification Accuracy: 37.80%

Model 3 (Motion blur):
37/37 ━━━━━━━━━━━━━━━━━━━━ 0s 6ms/step
- Classification f1: 98.56%
- Classification Accuracy: 44.64%
```

The accuracy on the corrupted datasets is close to random and much smaller than on the clean dataset for the baseline model. The augmented model performes much better on the motion blured dataset than the baseline model.

Task 2.3 Discussion (no coding)

- Which noise types affect the model most?
- What techniques could improve robustness (data augmentation, adversarial training, denoising pre-processing, larger models)?

Motion blur seems to affect the model more than the gaussian noise.

Techniques like data augmentation, adversarial training and denoising pre-processing generally improve robustness.

Part 3: Results & Comparison

- Summarize results (table/plot):
  - Baseline (clean) vs. Augmented (if attempted)
  - Clean vs. each corrupted test set (Task 2.2)

```
from tabulate import tabulate

print("Accuracy of baseline- and augmented model:\n")

header = ["", "Baseline model", "Augmented model"]
left_column = ["Clean", "Gaussian noise", "Motion blur"]

data_table = [["Clean", accuracy_2, accuracy_3],
    ["Gaussian noise", accuracy_2_gauss, accuracy_3_gauss],
    ["Motion blur", accuracy_2_blur, accuracy_3_blur]]

data_table = [row[0:1]+[f"{el*100:.2f} %" for el in row if isinstance(el, float)] for row in data_table]

data_table_transpose = [[data_table[i][j] for i in range(len(data_table))] for j in range(len(data_table[0]))]
```

```
print(tabulate(
    data_table,
    headers=header,
    tablefmt="presto",
    colalign=("left", "center", "center")
    ))
```

```
Accuracy of baseline- and augmented model:

                 |  Baseline model  |  Augmented model
----------------+------------------+-------------------
 Clean           |     82.27 %      |      83.82 %
 Gaussian noise |     37.80 %      |      37.80 %
 Motion blur     |     36.85 %      |      44.64 %
```

- Brief discussion:
  - Which corruptions hurt most, and why?
  - Did augmentation help? Which transforms mattered?
  - What would you try next (e.g., stronger augmentation, adversarial training, denoising pre-processing, larger models, early stopping, ensembling)?

The motion blur seems to hurt more than the gaussian noise. This is because the model uses a convolution layer, then a pooling layer at the start.

Augmentation helped with the motion blurred images. This is beacaus the augmentation we used shifted, zoomed and rotated the images. We essentially made the model more resilient to the type of transformations that happens in motion